

FILTER DESIGN TEST CASE

TC-SAW (Temperature Compensated SAW) - Band 1 (2100 MHz) for IoT Narrowband Module
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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Priority:	P3 - Medium

1. OBJECTIVE

This document describes the design, simulation, and characterization of a TC-SAW (Temperature Compensated SAW) filter targeting Band 1 (2100 MHz) for IoT Narrowband Module. The filter is designed on 36° YX LiTaO3 substrate with a target center frequency of 2314.5 MHz and bandwidth of 362.4 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the IoT Narrowband Module module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: TC-SAW (Temperature Compensated SAW)
- Frequency Band: Band 1 (2100 MHz)
- Application: IoT Narrowband Module
- Substrate: 36° YX LiTaO3
- Package: WLP (Wafer Level Package)
- Design Tool: Ansys HFSS 2024R1
- Design Center: Irvine Design Center

This specification applies to all variants of the TC-SAW (Temperature Compensated SAW) design targeting Band 1 (2100 MHz). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	2314.5 MHz
Bandwidth (3 dB):	362.4 MHz
Insertion Loss Target:	<= 1.4 dB
Return Loss Target:	>= 20.0 dB
Out-of-Band Rejection:	>= 36.0 dB
Isolation (Duplexer):	>= 36.0 dB
Q Factor:	3471
Temperature Coefficient:	-11.4 ppm/°C
Die Size:	2.8 x 0.8 mm
Wafer Size:	6-inch
Number of Resonators:	7
Electrode Material:	Al

Passivation: Si3N4

4. TEST PROCEDURE

1. Open Ansys HFSS 2024R1 and load the TC-SAW (Temperature Compensated SAW) template project.
2. Define the substrate stack: 36° YX LiTaO3 with specified layer thicknesses.
3. Set design targets: center freq 2314.5 MHz, BW 362.4 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 1.4 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (WLP (Wafer Level Package)).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (6-inch, 36° YX LiTaO3).
10. Perform on-wafer probing using Rohde & Schwarz ZNA67.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 6944 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 1.4 dB
- Return loss in passband shall be >= 20.0 dB
- Out-of-band rejection at +/- 544 MHz offset >= 36.0 dB
- Group delay variation across passband shall be <= 8 ns
- Temperature drift over -40°C to +85°C shall be <= 3.3 MHz
- Power handling shall be >= +27 dBm at 1 dB compression
- ESD tolerance >= 500 V (HBM)
- Die yield on qualification lot >= 85%

6. TEST RESULTS

Measurement Date: 2025-06-19
Devices Tested: 88
Insertion Loss (meas): 1.05 dB
Return Loss (meas): 21.5 dB
Rejection (meas): 38.9 dB
Isolation (meas): 40.8 dB
Q Factor (meas): 3471
Group Delay Variation: 7.0 ns
Power Handling: +36 dBm
Die Yield: 95.9%

Result Classification: NEEDS REVIEW

7. OBSERVATIONS AND FINDINGS

- a) The TC-SAW (Temperature Compensated SAW) design demonstrated below-target performance for Band 1 (2100 MHz).
- b) Insertion loss met target specification.
- c) Out-of-band rejection exceeded the minimum requirement.
- d) Temperature stability was within specification across full range.
- e) Batch-to-batch reproducibility confirmed over 3 wafer lots.

8. CORRECTIVE ACTIONS

- 1. Review near-band rejection margin for worst-case temperature.
- 2. Re-run EM simulation with refined mesh for better accuracy.
- 3. Schedule follow-up measurement after design tweak.

9. SIGN-OFF

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Approver: Dr. James Liu Date: 2025-07-03

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